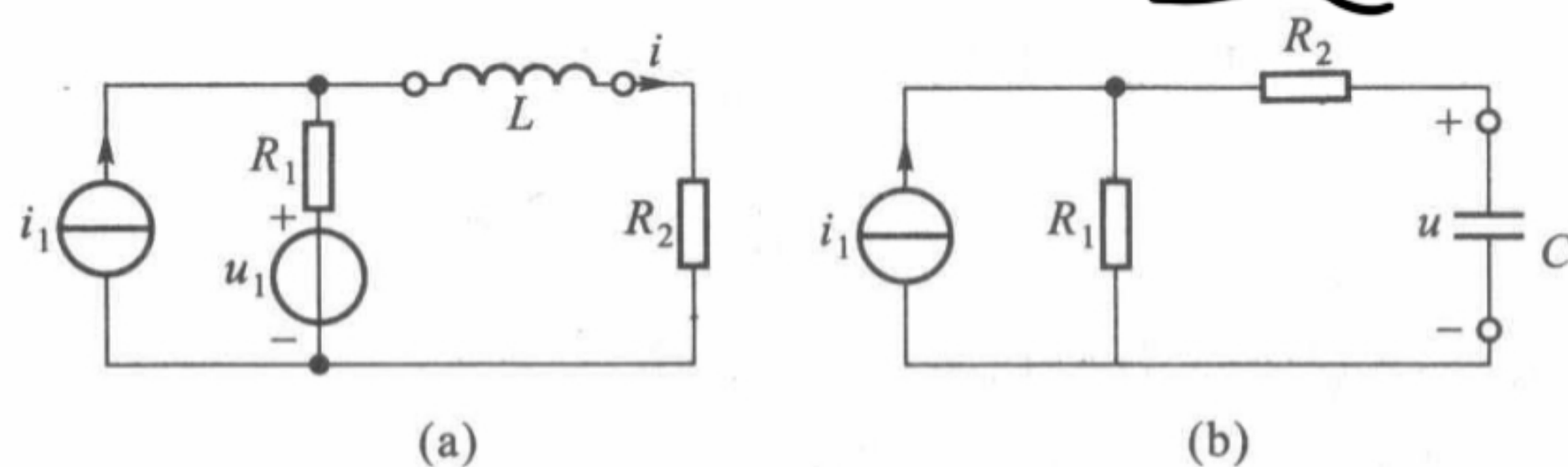


电分作业第六章

6-1 图题 6-1 两电路中, $R_1 = 100 \Omega$, $R_2 = 200 \Omega$, $R_3 = 300 \Omega$, $L = 2 \text{ mH}$, $C = 1 \mu\text{F}$ 。

- (1) 把各电路中除动态元件以外的部分化简为戴维南或诺顿等效电路;
- (2) 利用化简后的电路列出图中所注明输出量 u 或 i 的微分方程。



图题 6-1

6-1 (1)

(a) $u_{oc} = i_1 R_1 + u_1 = 100i_1 + u_1$
 $r = R_1 + R_2 = 300 \Omega \Rightarrow$

(b) $\Rightarrow$ $\Rightarrow$

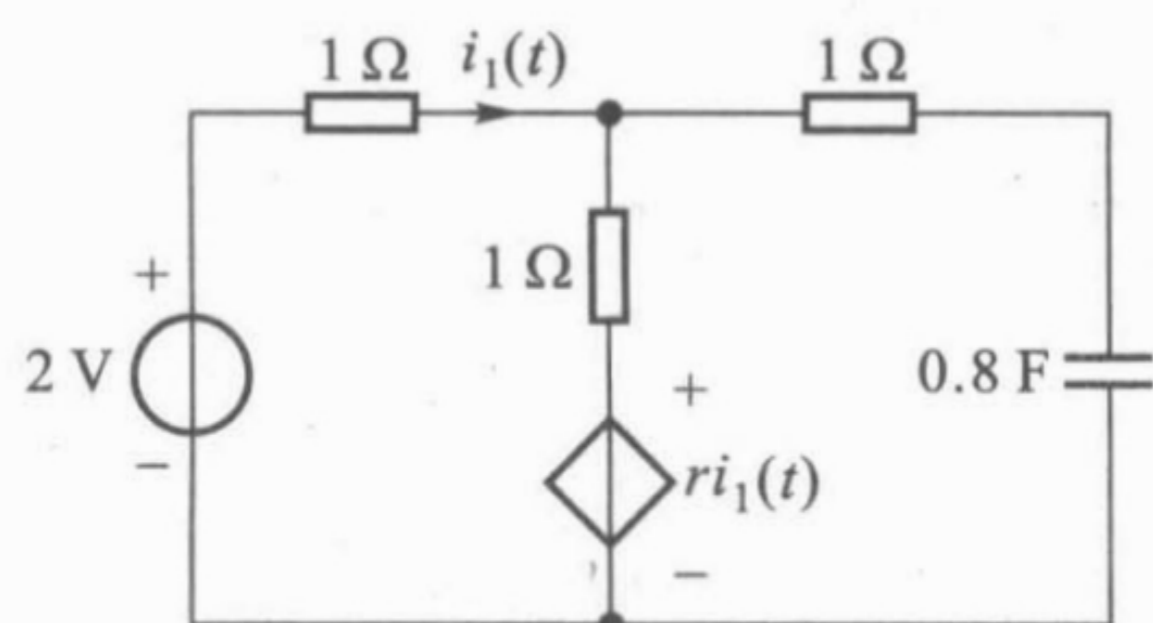
(2)

(a) $\Rightarrow$ KVL: $L \frac{di}{dt} - 100i_1 - u_1 + 300i = 0$
 $\Leftrightarrow L \frac{di}{dt} + 300i = 100i_1 + u_1$
 $\Leftrightarrow 2 \times 10^{-3} \frac{di}{dt} + 300i = 100i_1 + u_1$

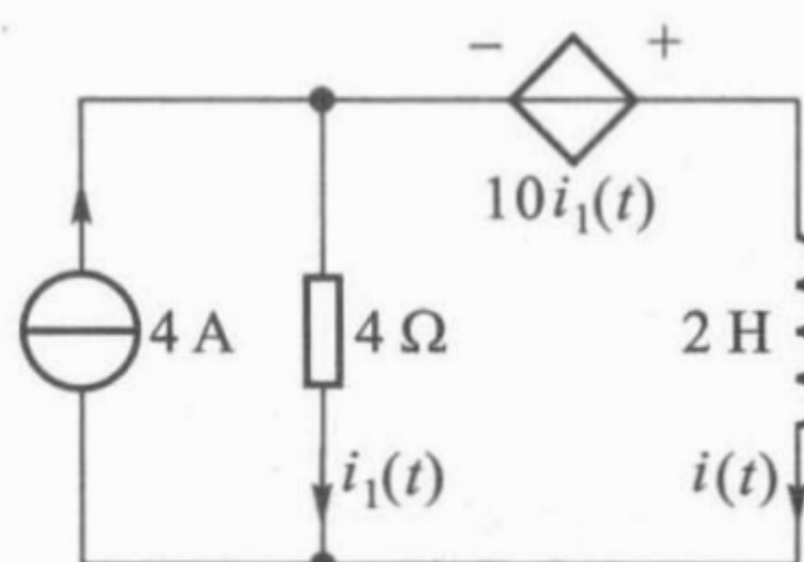
(b) $\Rightarrow$ KVL: $u - 100i_1 + 300 \cdot C \frac{du}{dt} = 0$ $\Leftrightarrow 300C \frac{du}{dt} + u = 100i_1$
 $\Leftrightarrow 3 \times 10^{-4} \frac{du}{dt} + u = 100i_1$

图题 6-7 所示电路, 电压源于 $t=0$ 时开始作用于电路, 试求 $i_1(t)$, $t \geq 0$ 。已知受控源参数 $r=2 \Omega$ 。

6-9 电路如图题 6-8 所示, 电源于 $t=0$ 时开始作用于电路, 试求 $i(t)$, $t \geq 0$ 。

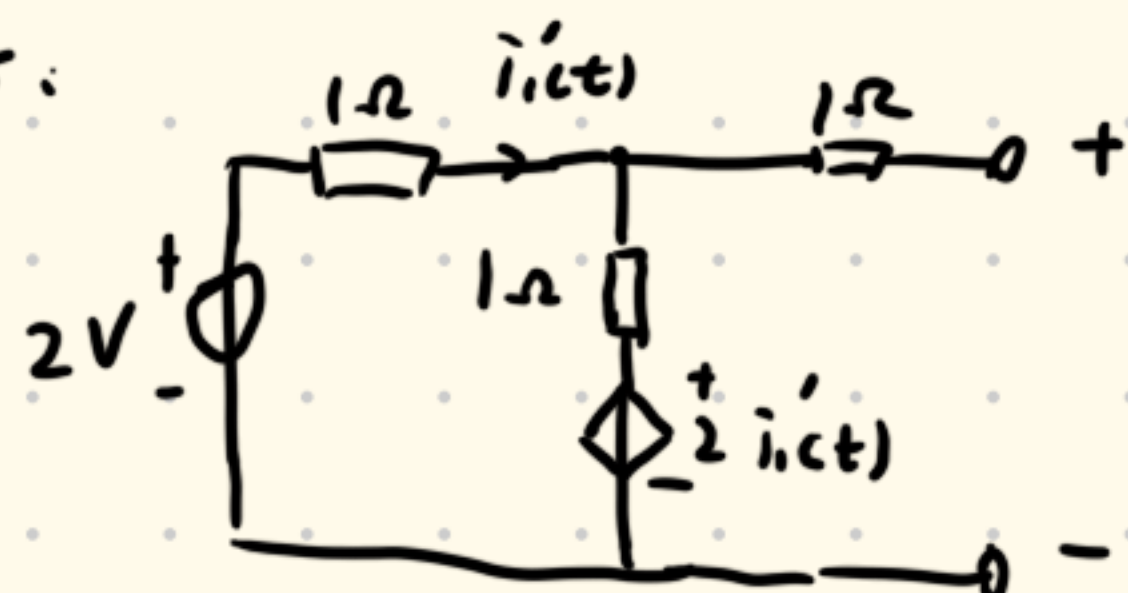


图题 6-7

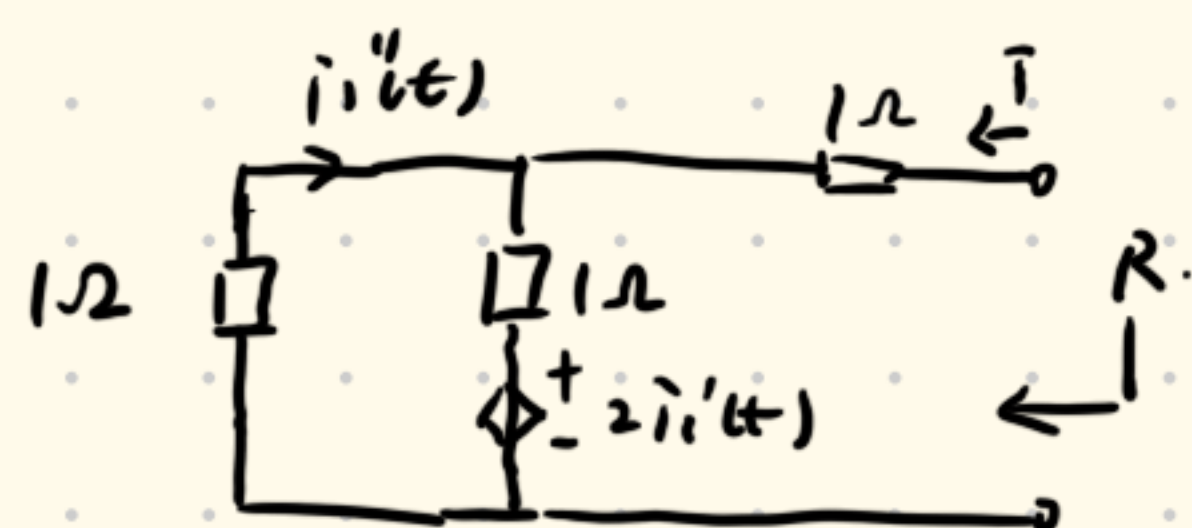


图题 6-8

6-8:

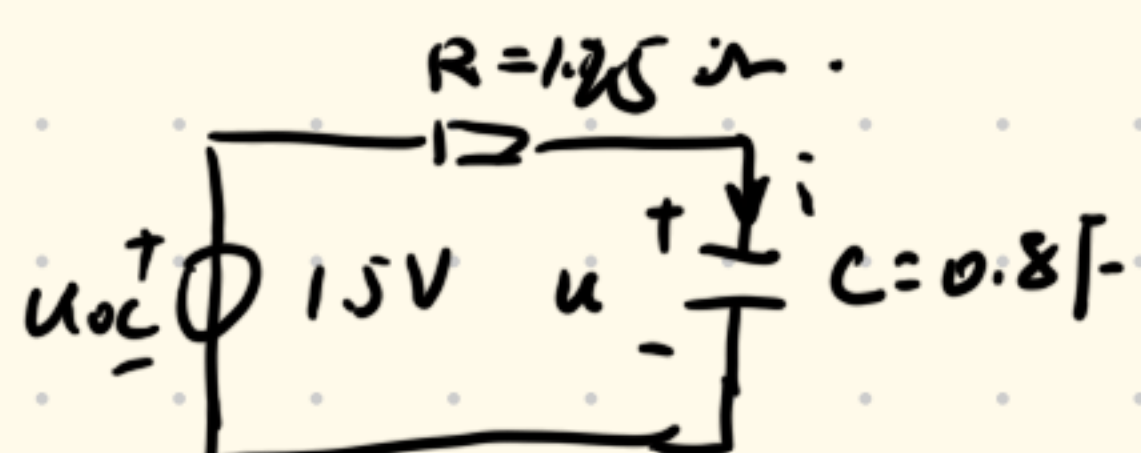


$$u_{oc} = 3i_1'(t) \quad \begin{cases} 2i_1'(t) + 2i_1'(t) - 2 = 0 \end{cases} \Rightarrow \begin{cases} u_{oc} = 1.5V \\ i_1'(t) = 0.5A \end{cases}$$



$$u = i - i_1''(t) \quad \begin{cases} -i_1''(t) = i + i_1''(t) + 2i_1''(t) \end{cases} \Rightarrow \begin{cases} u = \frac{5}{4}i \\ i_1''(t) = -\frac{1}{4}i \end{cases} \Rightarrow R = \frac{u}{i} = \frac{5}{4}\Omega$$

戴维南等效电路:



KVL 方程: $R \cdot C \cdot \frac{du}{dt} + u = u_{oc}$

$$\tau = CR = 1s$$

$$\Rightarrow u_c(t) = 1.5(1 - e^{-t}) = \frac{3}{2}(1 - e^{-t})$$

$$i_c(t) = C \cdot \frac{du}{dt} = \frac{6}{5}e^{-t}$$

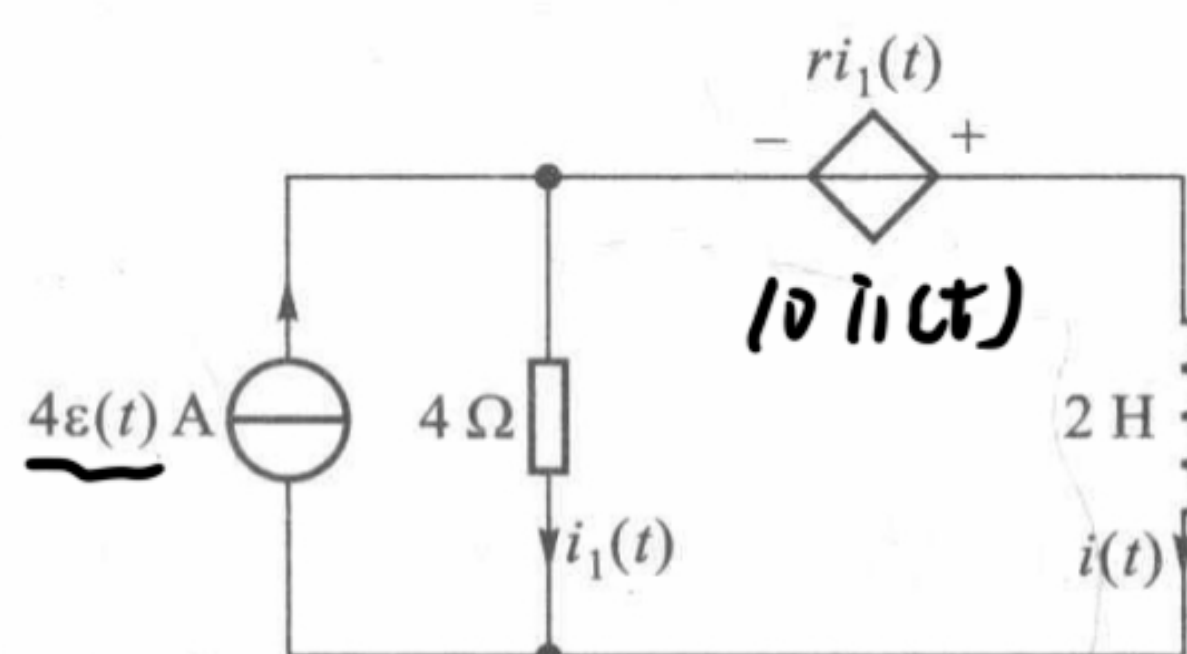
$$2: 1 \cdot i_c(t) + u_c(t) = 1 \cdot [i_1(t) - i_c(t)] + 2i_1(t)$$

$$\Rightarrow \frac{1}{5}e^{-t} + \frac{3}{2}(1 - e^{-t}) = 3i_1(t) - \frac{6}{5}e^{-t}$$

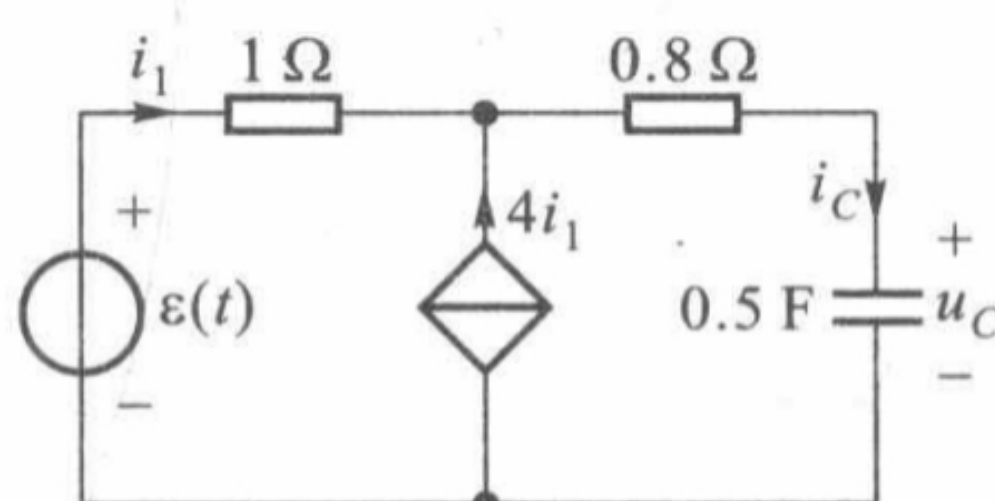
$$\Rightarrow \frac{12}{5}e^{-t} + \frac{3}{2} - \frac{3}{2}e^{-t} = 3i_1(t) \Rightarrow \underline{i_1(t) = (\frac{4}{5} - \frac{1}{2})e^{-t} + \frac{1}{2}} = \underline{0.5 + 0.3e^{-t}}$$

6-16 图题 6-15 所示电路, 已知受控源参数 $r = 10\Omega$, 求阶跃响应 $i(t)$ 、 $i_1(t)$ 。

6-17 试求图题 6-16 所示电路的阶跃响应 $u_c(t)$ 、 $i_1(t)$ 。



图题 6-15



图题 6-16

$$KCL: 4\epsilon(t) = i_1(t) + i_c(t)$$

$$KVL: 10i_1(t) + 4i_1(t) - L \frac{di_c(t)}{dt} = 0$$

$$\Rightarrow 4\epsilon(t) = \frac{1}{7} \cdot \frac{di_c(t)}{dt} + i_c(t)$$

$$i_c(t) = 4\epsilon(t)(1 - e^{-7t})$$

$$i_1(t) = 4\epsilon(t) \cdot e^{-7t}$$

6-18 求 RC 串联电路在冲激电压源 $\delta(t)$ 作用下电流的冲激响应。

6-19 求 RL 并联电路在冲激电流源 $\delta(t)$ 作用下电压的冲激响应。



$$u_C(t) = \varepsilon(t) (1 - e^{-\frac{t}{RC}})$$

$$i_C(t) = C \frac{du_C(t)}{dt} = \frac{1}{R} e^{-\frac{t}{RC}} \cdot \varepsilon(t) \text{ A}$$

$$h(t) = \frac{di_C(t)}{dt} = \left[-\frac{1}{RC} e^{-\frac{t}{RC}} \cdot \varepsilon(t) + \frac{1}{R} e^{-\frac{t}{RC}} \delta(t) \right] \text{ A}$$

6-23 图题 6-20 所示电路中, 已知电感电压 $u_{cd}(0_+) = 18 \text{ V}$, 求 $u_{ab}(t), t \geq 0$ 。

6-24 (1) 如图题 6-21(a) 所示 RL 电路, 开关断开切断电源瞬间开关将出现电弧, 试加以解释。

$$R_0 = 9 + (3+1) \parallel (4+8) = 12 \Omega$$

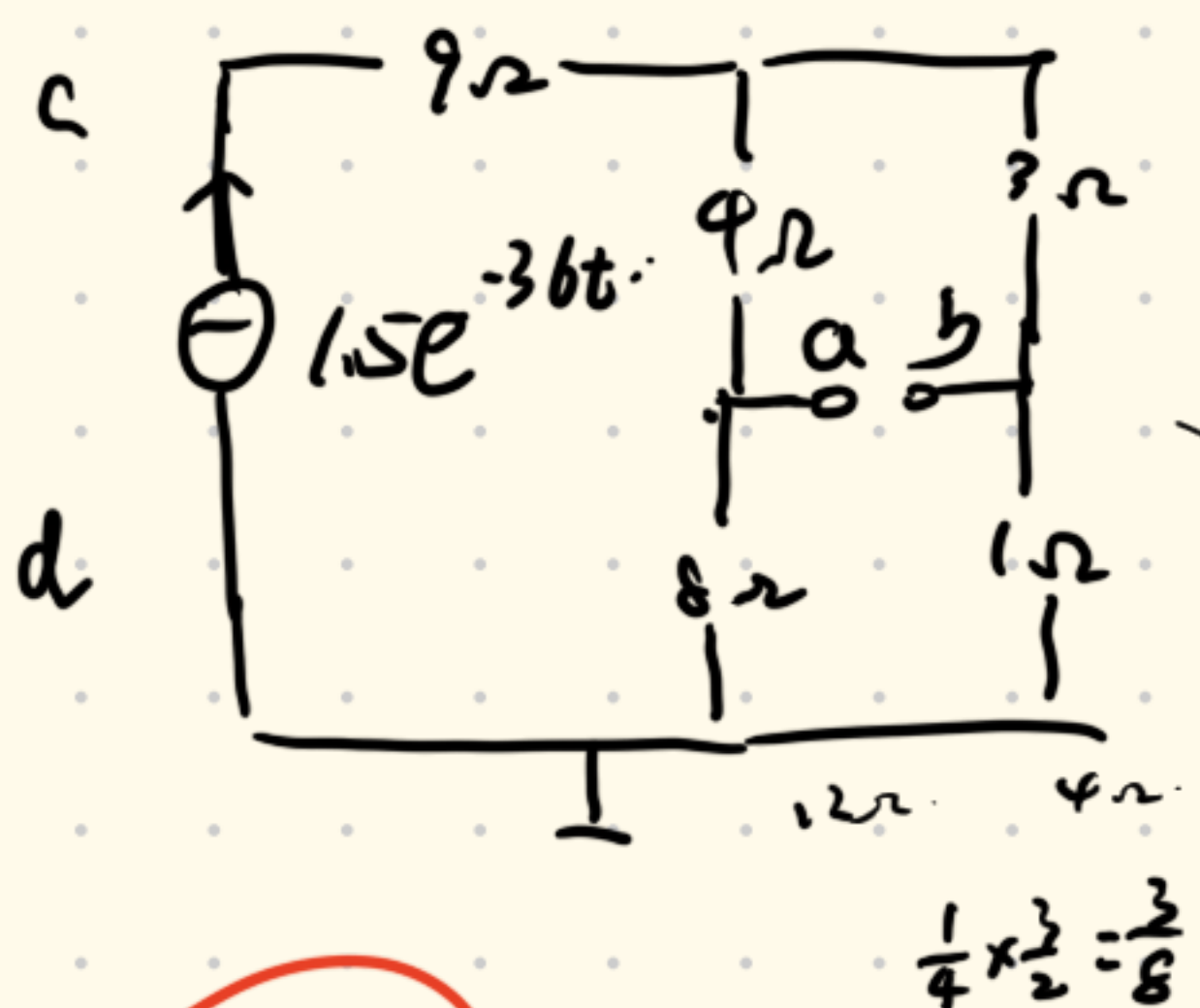
$$\tau = \frac{L}{R} = \frac{\frac{1}{3}}{12} = \frac{1}{36} \text{ s}$$

$$i_{cd}(t) = i_0 e^{-36t}$$

$$u_{cd}(t) = L \frac{di_{cd}(t)}{dt} = -12 i_0 e^{-36t}$$

$$u_{cd}(0) = 18 \text{ V} \Rightarrow i_0 = -1.5 \text{ A}$$

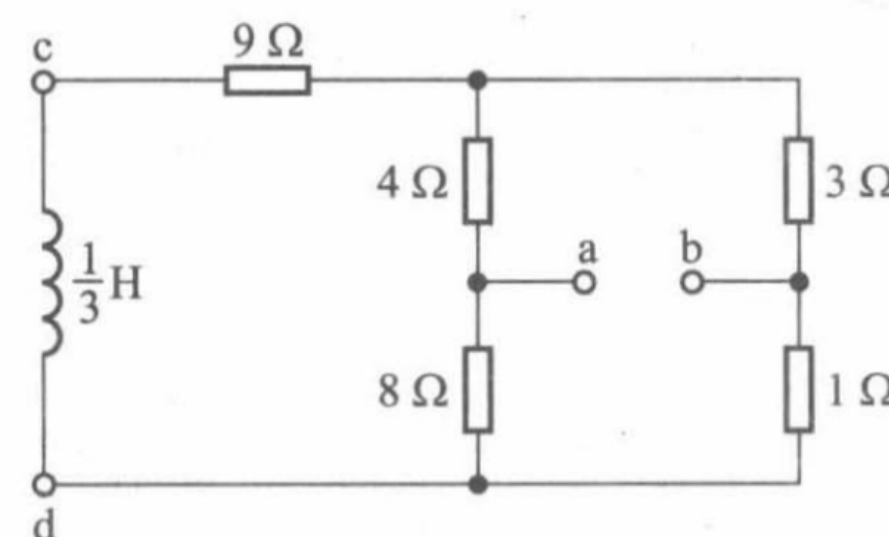
∴ 将电感量换成电流源。



$$u_a = \frac{1}{4} \times \frac{3}{2} e^{-36t} \times 8 = 3 e^{-36t}$$

$$u_b = \frac{3}{4} \times \frac{3}{2} e^{-36t} \times 1 = \frac{9}{8} e^{-36t}$$

$$\therefore u_{ab} = u_a - u_b = \frac{5}{8} e^{-36t} \text{ V } (t \geq 0)$$



图题 6-20

6-23 电路如图题 6-29 所示, 若电路初始状态为零, $i_s(t) = 4\varepsilon(t) \text{ A}$ 时, $i_L(t) = 2(1 - e^{-t})\varepsilon(t) \text{ A}$,

$u_R(t) = (2 - 0.5e^{-t})\varepsilon(t) \text{ V}$ 。试求 $i_L(0) = 2 \text{ A}$, $i_s(t) = 2\varepsilon(t) \text{ A}$ 时, $i_L(t)$ 和 $u_R(t), t > 0$ 。

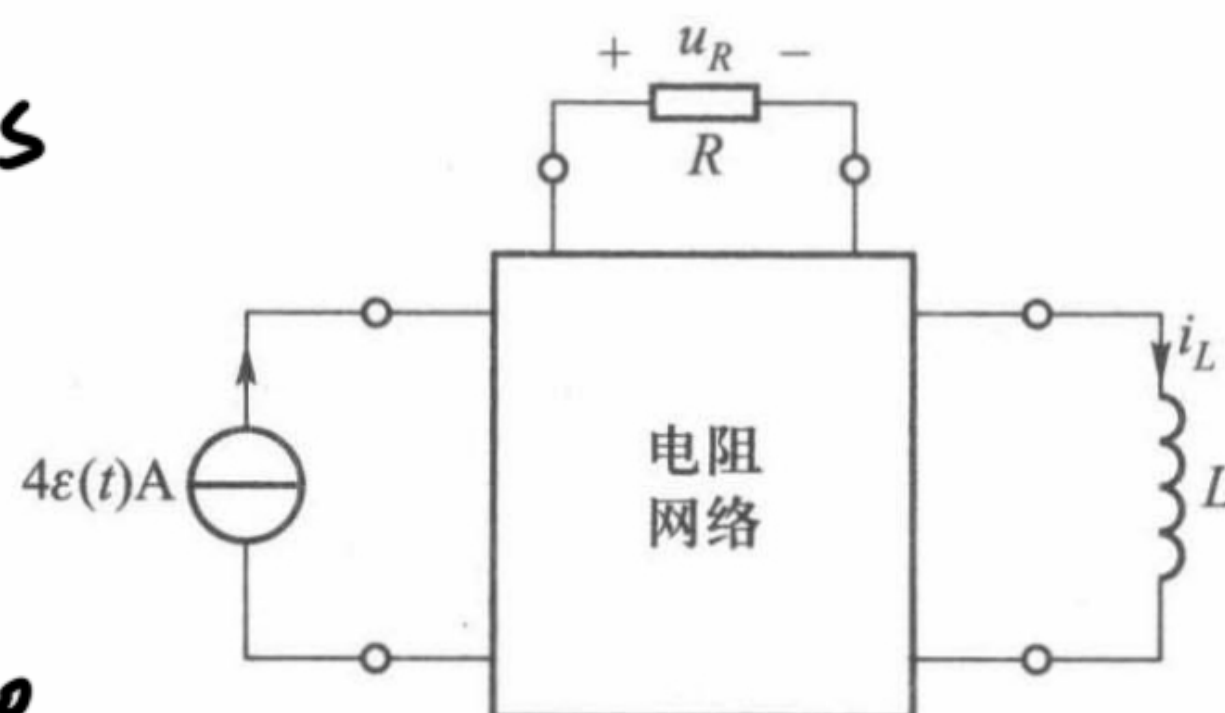
① 求 $i_L(t)$

零状态: $i_L'(t) = \frac{1}{2} (1 - e^{-t}) \cdot 2\varepsilon(t) = (1 - e^{-t})\varepsilon(t) \text{ A} \quad \tau = 1 \text{ s}$

零输入: $i_L(0) = 2 \text{ A}, i_s(t) = 2\varepsilon(t) \text{ A}$

$$i_L''(t) = 2e^{-t}\varepsilon(t) \text{ A}, t > 0$$

$$\therefore i_L(t) = i_L'(t) + i_L''(t) = (1 + e^{-t})\varepsilon(t) \text{ A}, t > 0$$



图题 6-29

② 求 $u_R(t)$

零状态: $u_R'(t) = \frac{1}{2} (2 - 0.5e^{-t})\varepsilon(t) = (1 - \frac{1}{4}e^{-t})\varepsilon(t) \text{ V}$

零输入: 电感量换为 $2e^{-t} \text{ A}$ 电流源。

$$u_R''(t) = 1 + 2e^{-t}\varepsilon(t) \text{ A}$$

根据初始条件

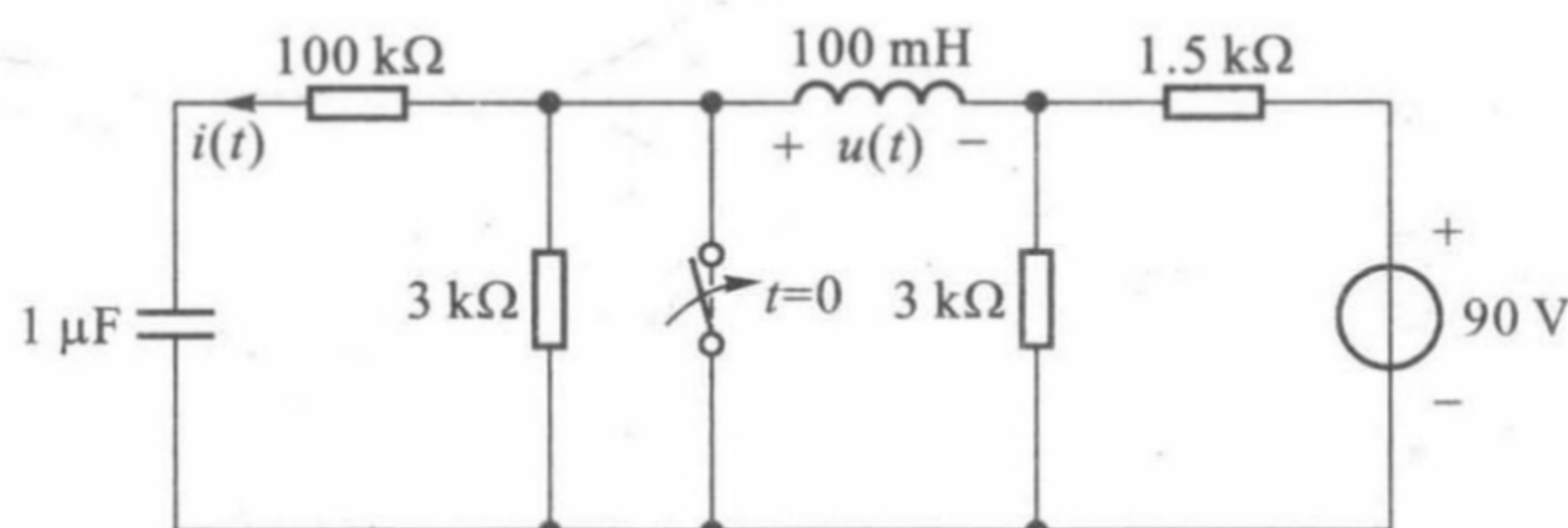
$$\varepsilon(t) \left[2 - \frac{1}{4} e^{-t} \right] = H_1 \cdot 4 \cdot \varepsilon(t) + H_2 \cdot 2(1 - e^{-t}) \varepsilon(t)$$

$$2 - \frac{1}{4} e^{-t} = 4H_1 + 2H_2 - 2H_2 e^{-t}$$

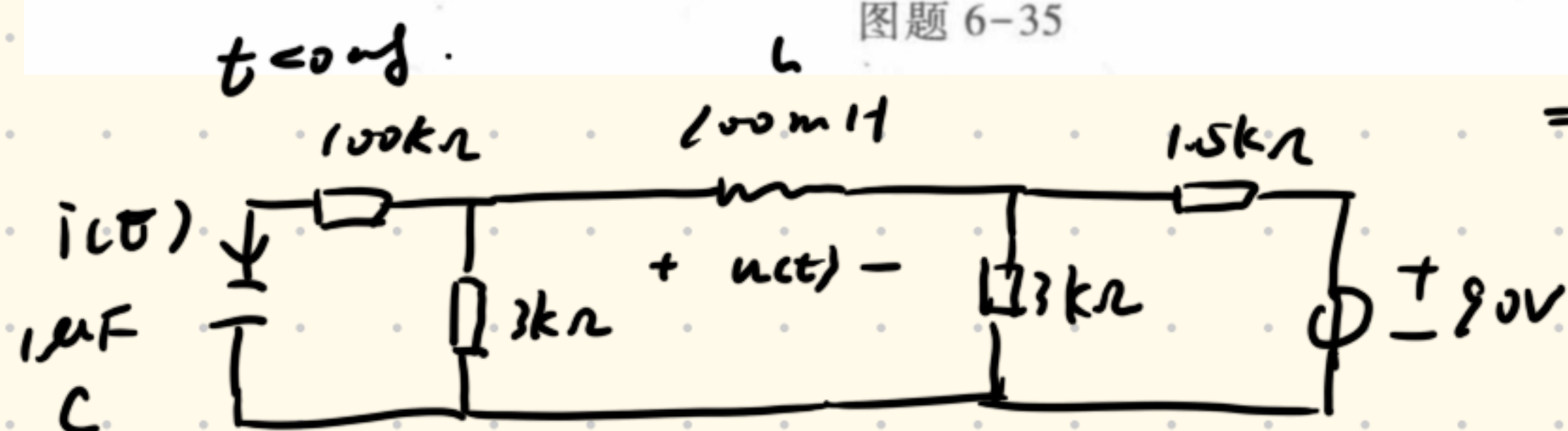
$$\Rightarrow \begin{cases} H_1 = \frac{1}{8} \\ H_2 = \frac{1}{4} \end{cases} \quad \therefore u_R(t) = \frac{1}{2} e^{-t} \varepsilon(t), \text{ V}$$

$$\therefore u_R(t) = u_R'(t) + u_R''(t) = \left(1 + \frac{1}{4} e^{-t} \right) \varepsilon(t) \text{ V}$$

6-41 图题 6-35 所示电路, 求 $i(t)$ 和 $u(t)$, $t \geq 0$ 。已知 $t=0_-$ 时电路已处于直流稳态。



图题 6-35



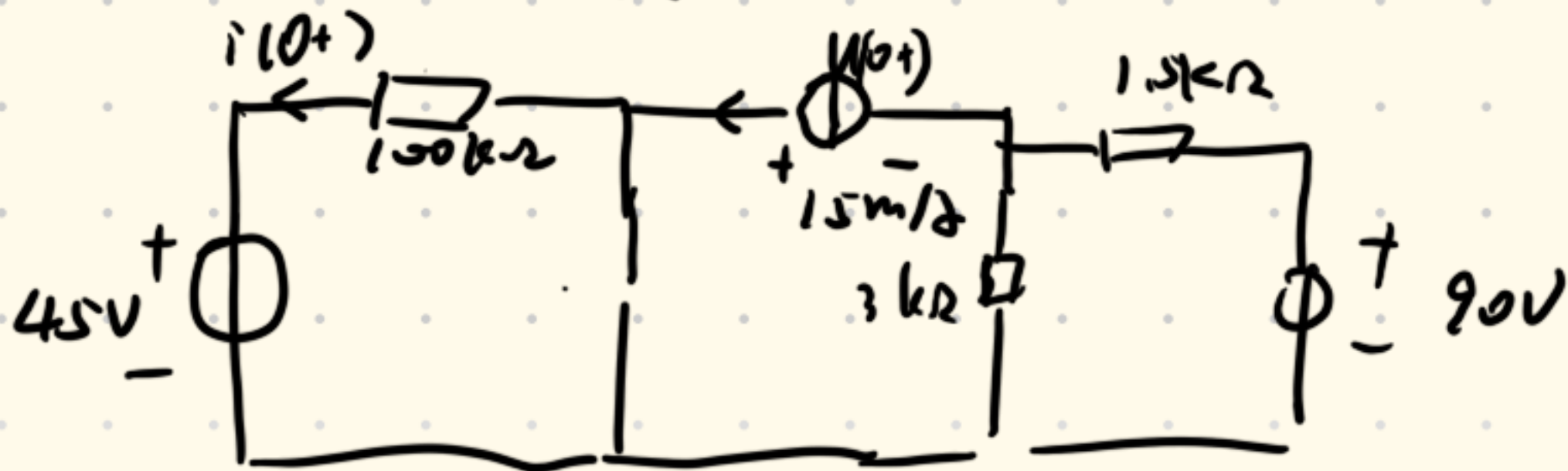
$$\begin{aligned} \frac{90\text{V}}{3\text{k}\Omega} \\ = 30 \times 10^{-3} \text{ A} \\ = 3 \times 10^{-2} \text{ A} \\ 1.5 \times 10^{-2} \text{ A} \end{aligned}$$

$$\frac{90}{1.5\text{k}\Omega}$$

$$u_C(0_-) = u_C(0_+) = 45\text{V}$$

$$i_L(0_-) = i_L(0_+) = -1.5 \times 10^{-2} \text{ A} = -15\text{mA}$$

$t = 0_+$ 时 等效电路:



$$i(0_+) = -\frac{45}{100 \times 10^3} = -0.45\text{mA}$$

$$u(0_+) = -90 \times \frac{2}{3} + \frac{3 \times 1.5}{3+1.5} \times 1.5 = -45\text{V}$$

$$i(\infty) = 0, \quad u(\infty) = 0$$

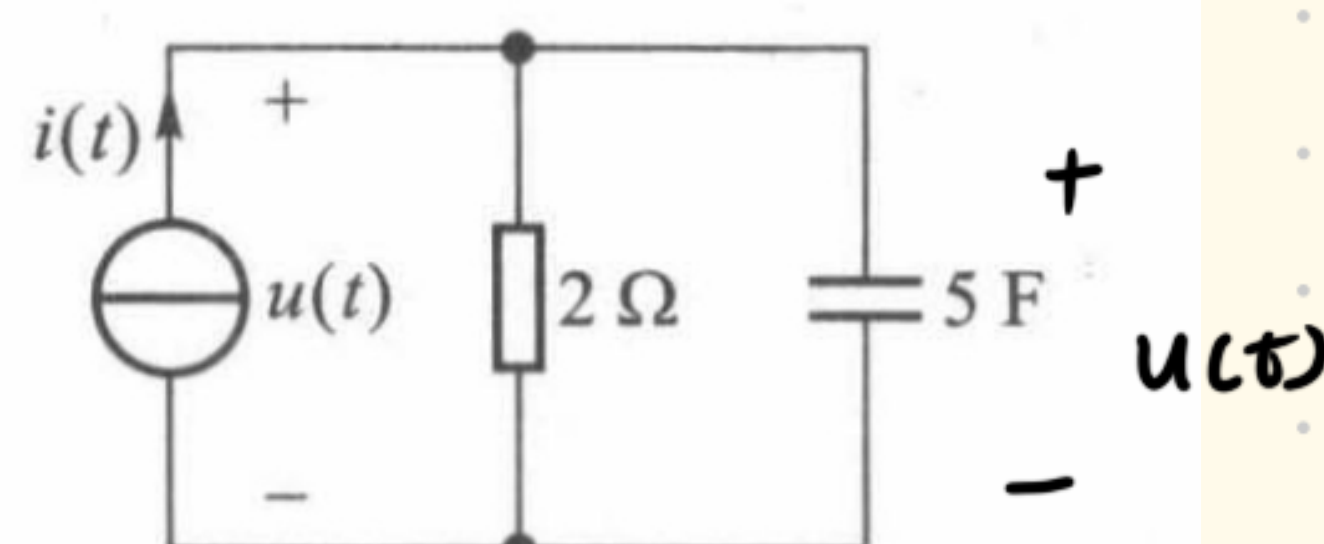
$$\tau_1 = RC = 100 \times 10^3 \times 10^{-6} = 0.1\text{s}$$

$$\tau_2 = \frac{L}{R} = \frac{100 \times 10^{-3}}{\frac{3 \times 1.5}{3+1.5} \times 10^3} = 10^{-4}\text{s}$$

$$i(t) = 0 + (-0.45 - 0)e^{-10t} = -0.45e^{-10t} \text{ mA} \quad t \geq 0$$

$$u(t) = 0 + (-45 - 0)e^{-10^4 t} = -45e^{-10^4 t} \text{ V} \quad t \geq 0$$

6-46 电路如图题 6-40 所示, 已知 $i(t) = 10 \text{ A}, t \geq 0; u(0) = 1 \text{ V}$ 。求电压 $u(t)$ 的零输入响应和零状态响应, 稳态响应和瞬态响应。



图题 6-40

$$\tau_1 = 2 \times 5 = 10 \text{ s}$$

零输入响应 $u_1 = 1 \cdot e^{-t} = e^{-0.1t}$

零状态响应: $u_2 = 20(1 - e^{-0.1t}) \text{ V}$

稳态响应: $u_1' = 20 \text{ V}$

瞬态响应: $u_2' = (1 - 20)e^{-0.1t} = -19e^{-0.1t} \text{ V}$

6-42.

$$u_0'(0_+) = u_0(\infty) = \frac{1}{2} \text{ V}$$

$$u_0'(\infty) = u_0(0_+) = \frac{5}{8} \text{ V}$$

$$\tau' = \frac{L}{R_0} = 1 \text{ s}$$

$$\therefore u_0'(t) = \frac{5}{8} + \left(\frac{1}{2} - \frac{5}{8}\right)e^{-t} = \left(\frac{5}{8} - \frac{1}{8}e^{-t}\right) \text{ V}, t \geq 0$$